

Khumba Lunganlung

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Technical Skills

Programming: C++, C, Python, JavaScript, TypeScript

Core: Object-Oriented Programming, Data Structures & Algorithms, Numerical Methods, Linear Algebra, Scientific Computing

Tools: Git, GitHub, VS Code, CMake, Vite

Web: React, Node.js, Express.js, MongoDB, Socket.io

Concepts: RESTful APIs, JWT Authentication, Real-Time Communication, Full-Stack Development

Education

Savitribai Phule Pune University

MSc in Scientific Computing

Pune, Maharashtra

2025 – Present

Kohima Science College, Jotsoma

BSc (Honours) in Computer Science

Kohima, Nagaland

2021 – 2024

Projects

Numerical Computing Library | C++

Jan 2026 – July 2026

github.com/Old-Daoist/Numerical-Computing-in-C-plus

- Developed a modular C++ numerical-computing library covering matrix algebra, root-finding, and complex-number operations.
- Implemented Gaussian elimination, determinant and inverse calculations, and core matrix operations using object-oriented design.
- Designed an abstract `RootMethod` base class with Bisection, Fixed Point, and Newton-Raphson implementations, using `std::function` for derivative support.
- Implemented a custom `myComplex` class with arithmetic, conjugate, and norm operations.

Five or More – Deep Reinforcement Learning Game | Python, Rust / Team Project

2026 – Present

github.com/Kunal-Kamod25/Five-Or-More-With-AI

- Contributed to the development of a Five or More game environment intended for training and evaluating Deep Reinforcement Learning agents.
- Developed and tested game-engine components and board-state mechanics using Rust with a modular project structure.
- Worked on the computational game logic required to represent, update, and evaluate game states for reinforcement-learning experiments.

Dirt Rally – Board Game | TypeScript, React, Tailwind CSS, Vite

Sept 2025 – Present

github.com/Old-Daoist/Dirt-Rally-Board-game

- Developed a multiplayer browser board game with a custom game engine supporting movement, tile resolution, combat, and obstacle logic.
- Implemented a time-seeded Mulberry32 pseudo-random number generator and typed game-state logic for deterministic game behaviour.
- Built PvP and PvCPU modes with CPU turn logic, lobby management, combat systems, action logging, and game-state handling.

Health-Hive | JavaScript, React, Node.js, Express, MongoDB, Socket.io

Aug 2025 – May 2026

github.com/Old-Daoist/Health-Hive

- Developed a full-stack health discussion platform with role-based access for users, verified doctors, and administrators.
- Implemented JWT authentication, email verification, password reset, and bcrypt password hashing with Mongoose-based data models.
- Built real-time messaging and notifications using Socket.io and administrative workflows for moderation and doctor verification.